ChemCritic 文献证据卡

PMC5225227 | Negishi 偶联 | 采集状态 imported_unchecked

标题	A triazine-based Ni(II) PNP pincer complex as catalyst for Kumada – Corriu and Negishi cross-coupling reactions
作者	Mastalir M., Kirchner K.
出处	Monatshefte Fur Chemie 2016
DOI	10.1007/s00706-016-1878-4
原文链接	https://pmc.ncbi.nlm.nih.gov/articles/PMC5225227/
许可	https://creativecommons.org/licenses/by/4.0/ Open AccessThis article is distributed under the terms of the Creative Comm

摘要（原文节选）

AbstractKumada – Corriu and Negishi cross-coupling reactions, catalyzed efficiently by a Ni(II) PNP pincer complex containing a triazine backbone, are described. The catalyst is able to react with both activated and inactivated aryl halides including chlorides as well as phenol derivatives such as tosylates and mesylates to give the corresponding cross-coupling products in good to excellent isolated yields. A high diversity of substrates was tested under moderate conditions for both types of reactions.Graphical Abstr ⋯

数据表 · Table 1（4 行，含数值行 4 行）

Catalyst screening for the Kumada – Corriu cross coupling

Entry	Catalyst	Base	Yield/%a
1	1	PhMgBr	47
2	2	PhMgBr	91
3	3	PhMgBr	86
4	4	PhMgBr	96

脚注（逐字）：4-Bromotoluene (1 mmol) and catalyst (2 mol%) in 3 cm3 THF, PhMgBr (1.3 mmol) was added and stirred at r.t. for 6 h aYield of pure isolated product after column chromatography

正文对应句（逐字）：The precatalysts 1 – 4 (Scheme 2) were prepared by treatment of NiCl2 · 6H2O with the respective PNP ligands as reported previously [27, 32] and are thermally robust and air-stable compounds. The reaction of 4-bromotoluene with phenyl magnesium bromide and phenyl zinc bromide in THF at room temperature

数据表 · Table 2（4 行，含数值行 4 行）

Catalyst screening for the Negishi cross coupling

Entry	Catalyst	Base	Yield/%a
1	1	PhZnBr	52
2	2	PhZnBr	92
3	3	PhZnBr	90
4	4	PhZnBr	94

脚注（逐字）：4-Bromotoluene (1 mmol) and catalyst (2 mol%) in 3 cm3 THF, PhZnBr (0.5 M in THF, 1.3 mmol) and LiCl (1.3 mmol) were added at r.t. and the solution was stirred at 60 ° C for 6 h aYield of pure isolated product after column chromatography

正文对应句（逐字）：The precatalysts 1 – 4 (Scheme 2) were prepared by treatment of NiCl2 · 6H2O with the respective PNP ligands as reported previously [27, 32] and are thermally robust and air-stable compounds. The reaction of 4-bromotoluene with phenyl magnesium bromide and phenyl zinc bromide in THF at room temperature

本卡为 NCBI PMC 开放获取 XML
的逐字摘录（标题、作者、表格行列、脚注、正文句均未改写），用于参照集候选与覆盖面登记；未逐条人工核验，标记 imported_unchecked，不写入判定记录、不参与 ChemCritic 引擎的可比性判定。对外引用一律回到 DOI 原文核对页码与表号。